First Order Q8

$$y' + y = e^{2x}$$

$\rightarrow y' + P(x)y = Q(x)$

$\rightarrow P(x) = 1, \; Q(x) = e^{2x}$

$$\mu(x) = e^{\int P(x)dx} = e^{\int dx} = e^{x}$$

$\rightarrow \overset{\times e^{x}}{e^{x}y' + e^{x}y} = e^{3x} \Rightarrow \frac{d}{dx}e^{x}y = e^{3x}$

$\rightarrow e^{x}y = \int e^{3x}dx = \frac{1}{3}e^{3x} + C$

$$y = \frac{1}{3}e^{2x} + Ce^{-x} \; \#$$

Second Order Q10

$$y'' - y = e^{x} + x$$

Slove Homogeneous Solution

Characteristic Eq $\rightarrow r^2 - 1 = 0 \; ; r = \pm 1$

Soln

$\Rightarrow y_h = C_1 e^{x} + C_2 e^{-x}$

Find Particular Solution, y_p

Let $F(x) = e^x + x \rightarrow y_p = y_{p_1} + y_{p_2}$

$u = Ax$

$v = e^x$

$\frac{d}{dx} uv = u'v + uv'$

$= Ae^x + Axe^x$

Slove y_{p1} for $e^x \Rightarrow y_{p_1} = Ae^x \cdot x$

$y'_{p_1} = A(e^x + xe^x)$

$y''_{p_1} = A(e^x + e^x + xe^x) = A(2e^x + xe^x)$

$y'' - y = e^x$

$A(2e^x + xe^x) - (Axe^x) = e^x$

$2Ae^x = e^x$

$A = \frac{1}{2} \Rightarrow y_{p_1} = \frac{1}{2}e^x x$

y_{p2} for x

$y_{p2} = Bx + C$

$y'_{p2} = B$

$y''_{p2} = 0$

$0 - Bx - C = x$

$(B+1)x = C$

$B = -1$

$C = 0$

$y_{p2} = -x$

$y_p = \frac{1}{2}xe^x - x$

$y = C_1 e^x + C_2 e^{-x} + \frac{1}{2}xe^x - x$ ※